

D3 Scale Hierarchy: Δr_s Across Physical Temperatures

Physical Context	Temperature [K]	Δr_s [m/bit]	$\Delta r_s / \ell_P$
Cosmic microwave background	2.725	4.31×10^{-66}	2.67×10^{-31}
Room temperature	300	4.74×10^{-65}	2.93×10^{-30}
Human body temperature	310	4.90×10^{-65}	3.03×10^{-30}
Hawking temp. (M☉)	6.17×10^{-8}	9.75×10^{-75}	6.03×10^{-40}
LHC quark-gluon plasma	5.5×10^{12}	8.70×10^{-55}	5.38×10^{-20}
Planck temperature	1.42×10^{32}	$2 \cdot \ln(2) \cdot \ell_P$	$2 \ln 2$ (exact)